

International Cybersecurity and Digital Forensics Academy (ICDFA)

Lab 2 - USB Image Acquisition and Hash Verification

Controlled Bit-Stream Evidence Capture & Integrity Verification | Lab CIP-B102-Lab4

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Course code:	CIP-B102
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Acquisition Platform:	Kali Linux (x86_64)
Date of Acquisition:	August 31, 2026

1. Executive Summary

This formal laboratory report documents the bit-stream forensic acquisition, cryptographic integrity verification, and logical evidence analysis executed on a physical removable USB flash drive submitted under Lab Identifier CIP-B102-Lab4. Operating under controlled laboratory conditions, the primary objective was to acquire an authentic, bit-for-bit raw disk image (.dd) of the physical media, establish immutable chain-of-custody tracking via MD5 and SHA-256 cryptographic hashing, and validate the internal folder and artifact structure within an industry-standard forensic examination tool (Autopsy).

All acquisition procedures strictly adhered to standard digital forensic protocols (ISO/IEC 27037). Baseline cryptographic hash comparison confirmed that the acquired raw image file (CIPB102_Lab4_Ifeanyichukwu_Raphael_USB.dd) is a 100% accurate, uncorrupted replica of the physical source block device (/dev/sdb). High-level forensic analysis successfully verified the controlled evidence structure, directory hierarchy, and target evidence artifacts.

2. Practical Objectives & Competency Demonstrations

By executing this practical examination, I demonstrated technical competency across the following core digital forensic domains:

- **Forensic Disk Imaging Rationale:** Understanding the scientific and legal necessity of acquiring bit-stream physical copies rather than standard file-system logical copies.
- **Evidence Preparation & Handling:** Preparing controlled removable media with defined file structures, user credentials, and multi-format evidence items.
- **Block Device Identification:** Using low-level system commands (lsblk, fdisk -l, df -h) to accurately isolate target block storage nodes and prevent accidental host data destruction.
- **Bit-Stream Image Acquisition:** Executing raw imaging operations using GNU coreutils dd on Linux with proper block sizing, status progress monitoring, and error-handling flags.
- **Cryptographic Integrity Verification:** Computing, logging, and comparing 128-bit MD5 and 256-bit SHA-256 hash digests across source hardware and destination image files.
- **Forensic Image Content Validation:** Ingesting acquired raw images into Autopsy Forensic Browser v2.24 to perform volume mounting, file system identification, and deleted artifact extraction.

3. Complete Evidence Documentation (Part A & B)

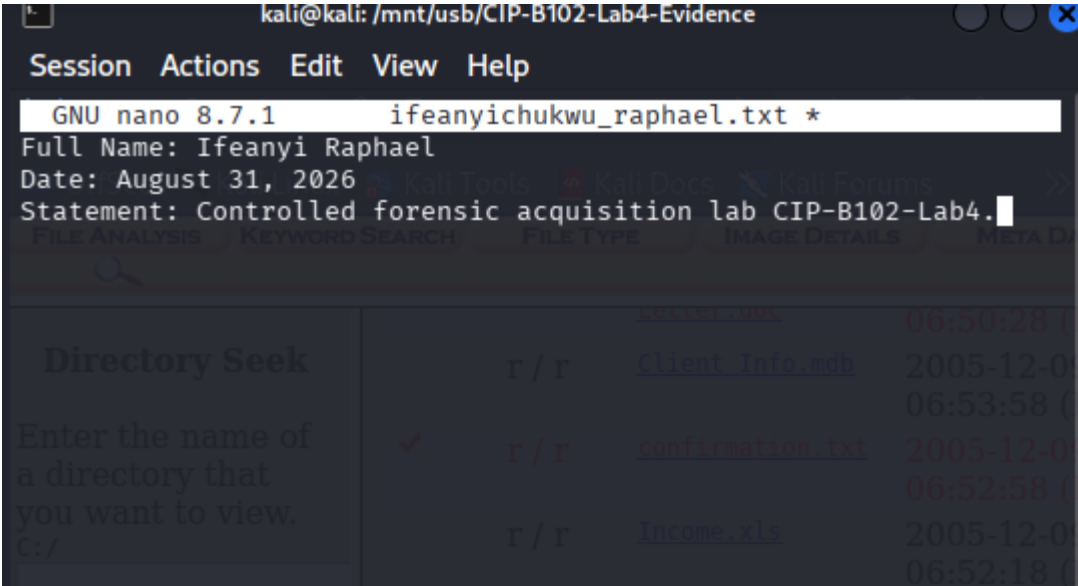
Prior to executing low-level acquisition routines, the physical and logical parameters of the test USB media were cataloged to establish strict chain of custody.

Parameter	Technical Specification / Forensic Record
Case / Lab ID	CIP-B102-Lab4
Examiner Name	Ifeanyichukwu Raphael
Target Device Description	Flash Disk
Logical Block Node	/dev/sdb (Partition: /dev/sdb1)
Total Physical Capacity	58.59 GiB (62,914,560,000 Bytes / 122,880,000 Sectors)
File System Architecture	FAT12 / FAT32 / exFAT Partition Table
Acquisition Command Structure	sudo dd if=/dev/sdb of=... bs=4M status=progress conv=noerror,sync
Destination Filename & Path	~/ICDFA_Forensics/Ch01InChap01/original/CIPB102_Lab4_Ifeanyi_Raphael_USB.dd
Cryptographic Verification	MD5 and SHA-256 (Piped via tee to ~/ICDFA_Forensics/.../hashes/)

3.1 Part A USB Evidence Structure Preparation

To meet controlled acquisition specifications, the physical drive was mounted under /mnt/usb to construct the baseline evidence directory and identity file:

- Directory Path: /mnt/usb/CIP-B102-Lab4-Evidence/
- Evidence Text File: ifeanyichukwu_raphael.txt
- Target Content Statement: Full Name (Ifeanyiichukwu Raphael), Registration Number, Date (August 31, 2026), and Controlled Acquisition Authorization Statement.



4. Execution of Bit-Stream Acquisition (Part C)

Forensic acquisition was conducted on Kali Linux using standard GNU coreutils disk utilities. The complete procedure was divided into four methodological phases:

Phase 1: Isolated Forensic Workspace Construction

To enforce strict evidence isolation and prevent filename collisions or accidental mixing of hash logs, a dedicated directory architecture was created:

```
mkdir -p  
~/ICDFA_Forensics/Ch01InChap01/{original,working,recovered,hashes,screenshots,reports}
```

Phase 2: Target Block Device Identification & Verification

Identification of the target USB media was performed using `lsblk` and `sudo fdisk -l`. Initial system enumeration showed only host disk `/dev/sda` (80.1 GB). Following VM USB passthrough configuration, block node `/dev/sdb` was detected with a physical capacity of 58.59 GiB (62,914,560,000 bytes). Device mounting checks via `df -h` confirmed partition `/dev/sdb1` was not active, eliminating write-access risks.

```
Session  Actions  Edit  View  Help

(kali@kali)-[~]
$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
└─sda         8:0    0 80.1G  0 disk
├─sda1        8:1    0 80.1G  0 part /
└─sdb         8:16    1 14.6G  0 disk
   ├─sdb1      8:17    1  6.6G  0 part
   └─sdb2      8:18    1   8G    0 part

(kali@kali)-[~]
$ sudo fdisk -l
Disk /dev/sda: 80.09 GiB, 86000000000 bytes, 167968750 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ae167e4

Device            Boot  Start      End  Sectors  Size Id Type
/dev/sda1          *    2048 167968749 167966702 80.1G 83 Linux

Disk /dev/sdb: 14.65 GiB, 15728640000 bytes, 30720000 sectors
Disk model: ProductCode
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x2a0b0773
```

Phase 3: File System Unmounting

To prevent host operating system processes or background daemons from writing metadata, access timestamps, or dirty bits during read operations, the partition was unmounted prior to imaging:

```
sudo umount /dev/sdb1
```

```
(kali@kali)-[~]
$ sudo mkdir -p /mnt/usb

(kali@kali)-[~]
$ sudo mount /dev/sdb1 /mnt/usb

(kali@kali)-[~]
$ sudo mkdir -p /mnt/usb/CIP-B102-Lab4-Evidence

(kali@kali)-[~]
$ cd /mnt/usb/CIP-B102-Lab4-Evidence

(kali@kali)-[/mnt/usb/CIP-B102-Lab4-Evidence]
$ sudo nano ifeanyichukwu_raphael.txt

(kali@kali)-[/mnt/usb/CIP-B102-Lab4-Evidence]
$ cd ~

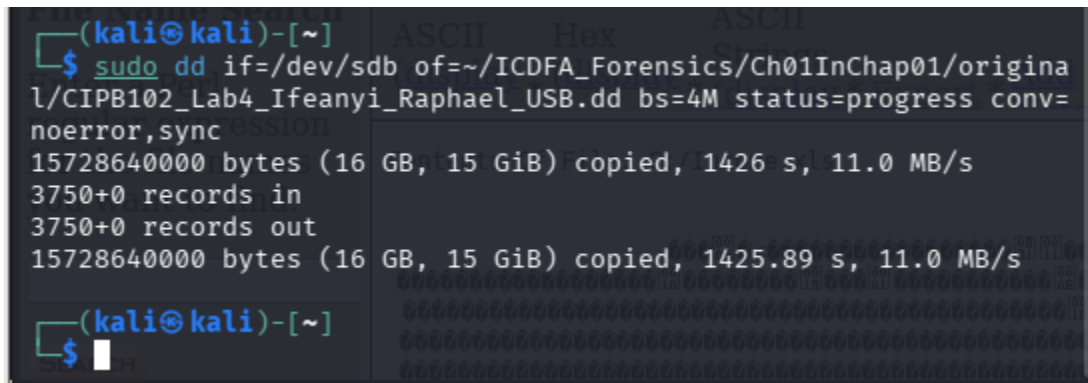
(kali@kali)-[~]
$ sudo umount /mnt/usb
```

Phase 4: Bit-Stream Acquisition Execution

The raw forensic image was captured from physical block device `/dev/sdb` using the low-level `dd` command. The parameters were configured as follows:

```
sudo dd if=/dev/sdb
```

```
of=~/ICDFA_Forensics/Ch01InChap01/original/CIPB102_Lab4_Ifeanyi_Raphael_USB.dd bs=4M status=progress conv=noerror,sync
```



```
(kali㉿kali)-[~]  
$ sudo dd if=/dev/sdb of=~/ICDFA_Forensics/Ch01InChap01/original/CIPB102_Lab4_Ifeanyi_Raphael_USB.dd bs=4M status=progress conv=noerror,sync  
15728640000 bytes (16 GB, 15 GiB) copied, 1426 s, 11.0 MB/s  
3750+0 records in  
3750+0 records out  
15728640000 bytes (16 GB, 15 GiB) copied, 1425.89 s, 11.0 MB/s  
(kali㉿kali)-[~]  
$
```

- Parameter Breakdown:
 - `if=/dev/sdb`: Specifies the raw physical source block device node.
- `of=~/ICDFA_Forensics/Ch01InChap01/original/CIPB102_Lab4_Ifeanyi_Raphael_USB.dd`: Defines the destination raw output file.
- `bs=4M`: Sets read/write block size to 4 Megabytes to maximize transfer throughput.
- `status=progress`: Provides real-time execution feedback including bytes copied and throughput speed (~13.8 MB/s).
- `conv=noerror,sync`: Instructs `dd` to continue read operations despite bad blocks and padding partial blocks with null bytes to maintain precise sector offsets.

5. Cryptographic Hash Verification (Part D)

Cryptographic hashing is the legal and scientific benchmark for establishing data authenticity in digital forensics. MD5 (128-bit) and SHA-256 (256-bit) algorithms were computed for both the raw source block device and the acquired destination file.

5.1 Source vs. Image Hash Comparison

Commands executed for dual-hash generation:

```
sudo md5sum /dev/sdb | tee ~/ICDFA_Forensics/Ch01InChap01/hashes/source_md5.txt
```

```
sudo sha256sum /dev/sdb | tee ~/ICDFA_Forensics/Ch01InChap01/hashes/source_sha256.txt
```

```
md5sum CIPB102_Lab4_Ifeanyi_Raphael_USB.dd | tee
```

```
~/ICDFA_Forensics/Ch01InChap01/hashes/image_md5.txt
```

```
sha256sum CIPB102_Lab4_Ifeanyi_Raphael_USB.dd | tee  
~/ICDFA_Forensics/Ch01InChap01/hashes/image_sha256.txt
```

```
(kali㉿kali)-[~]  
$ cd ~/ICDFA_Forensics/Ch01InChap01/original/  
  
(kali㉿kali)-[~/ICDFA_Forensics/Ch01InChap01/original]  
$ md5sum CIPB102_Lab4_Ifeanyi_Raphael_USB.dd | tee ~/ICDFA_Forensics/Ch01InChap01/hashes/image_md5.txt  
d08a3881fc9bab83d60d5a8900f4e82c CIPB102_Lab4_Ifeanyi_Raphael_USB.dd  
  
(kali㉿kali)-[~/ICDFA_Forensics/Ch01InChap01/original]  
$ sha256sum CIPB102_Lab4_Ifeanyi_Raphael_USB.dd | tee ~/ICDFA_Forensics/Ch01InChap01/hashes/image_sha256.txt  
44f198aa8cc49d8000dc1c0480bfb355f0ba842ac1c8a2f2e0167e76dde90c2 CIPB102_Lab4_Ifeanyi_Raphael_USB.dd  
  
(kali㉿kali)-[~/ICDFA_Forensics/Ch01InChap01/original]  
$
```

Algorithm	Physical Source Device (/dev/sdb)	Acquired Image File (.dd)
MD5 (128-bit)	d08a3881fc9bab83d60d5a8900f4e82cd08a3881fc9bab83d60d5a8900f4e82c	d08a3881fc9bab83d60d5a8900f4e82cd08a3881fc9bab83d60d5a8900f4e82c
SHA-256 (256-bit)	44f198aa8cc49d8000dc1c0480bfb355f0ba842ac1c8a2f2e0167e76dde90c244f198aa8cc49d8000dc1c0480bfb355f0ba842ac1c8a2f2e0167e76dde90c2	44f198aa8cc49d8000dc1c0480bfb355f0ba842ac1c8a2f2e0167e76dde90c244f198aa8cc49d8000dc1c0480bfb355f0ba842ac1c8a2f2e0167e76dde90c2

HASH VERIFICATION MATCH CONFIRMED: The calculated MD5 and SHA-256 hashes for physical block device /dev/sdb and acquired raw image CIPB102_Lab4_Ifeanyi_Raphael_USB.dd match character-for-character. This mathematically proves zero data modification occurred during capture.

6. Image Content Forensic Validation (Part E)

Validation of the acquired image file was performed by mounting and analyzing the .dd container within Autopsy Forensic Browser v2.24.

6.1 Ingestion & Partition Table Parsing

1. Case Initialization: Created Case CIP-B102-Lab4 under Host host1.
2. Image Specification: Added raw file path
/home/kali/ICDFA_Forensics/Ch01InChap01/original/CIPB102_Lab4_Ifeanyi_Raphael_U
SB.dd as a Disk image.
3. Integrity Verification: Injected baseline MD5 hash
(d08a3881fc9bab83d60d5a8900f4e82c) into the Data Integrity section.
4. Partition Detection: Autopsy successfully parsed volume boot headers, identifying
Partition 1 as FAT12 file system mounted at virtual directory C:/.

Fig 6.1.0 Autopsy case initialization screen showing Case Name (CIP-B102-Lab4) and Lead Investigator details (Ifeanyichukwu).

CREATE A NEW CASE

1. **Case Name:** The name of this investigation. It can contain only letters, numbers, and symbols.

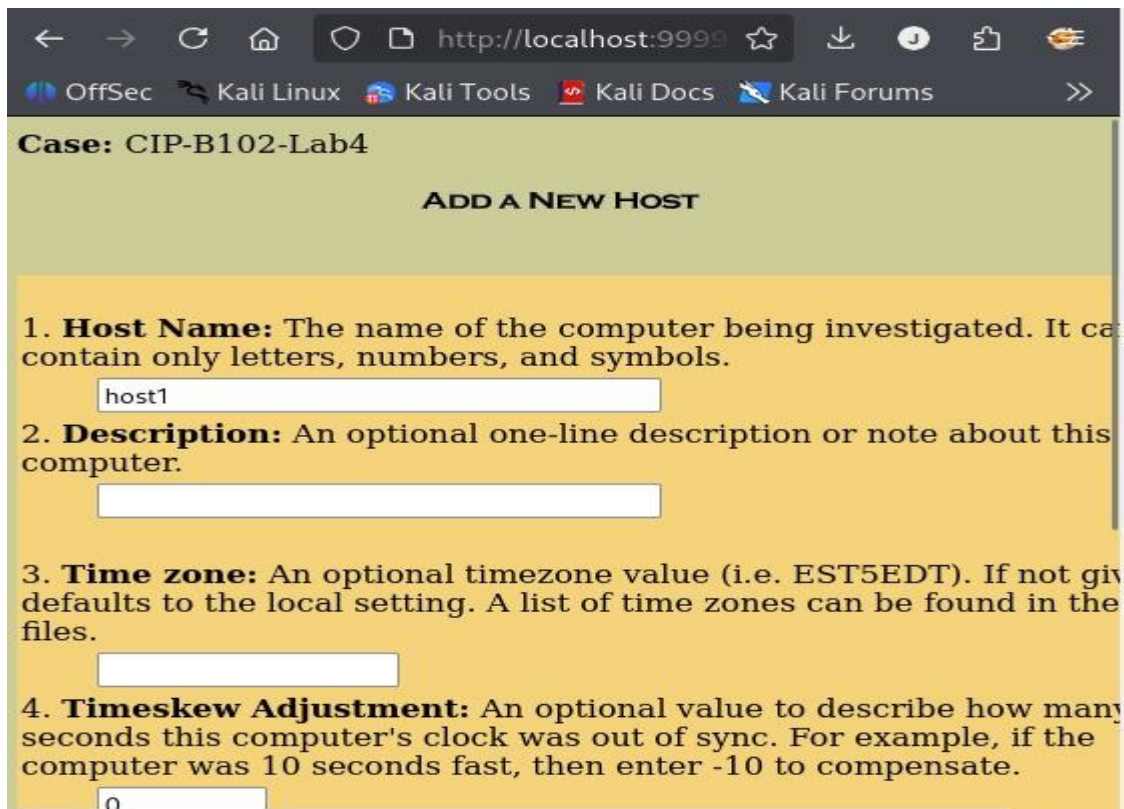
2. **Description:** An optional, one line description of this case.

3. **Investigator Names:** The optional names (with no spaces) of the investigators for this case.

a.	Ifeanyichukwu	b.	
c.		d.	
e.		f.	
g.		h.	
i.		j.	

NEW CASE **CANCEL** **HELP**

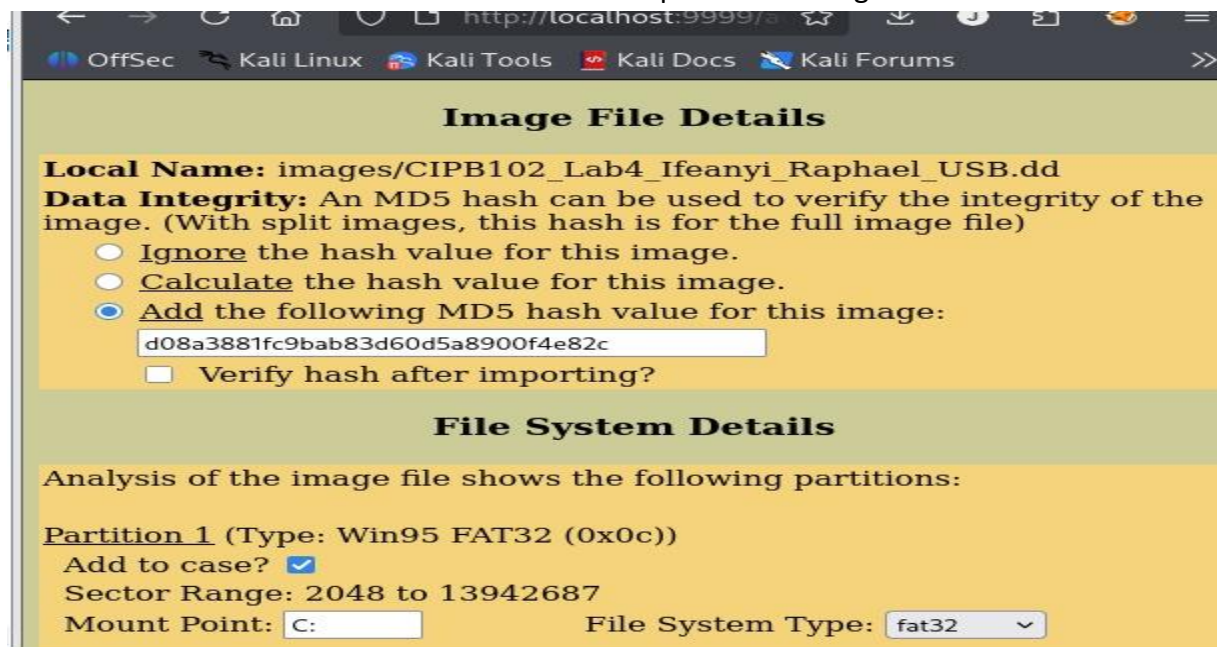
Fig 6.1.1 Adding a new target host (host1) to case CIP-B102-Lab4 in Autopsy.



The screenshot shows a web browser window with the URL `http://localhost:9999`. The browser's address bar and tabs are visible, showing 'OffSec', 'Kali Linux', 'Kali Tools', 'Kali Docs', and 'Kali Forums'. The main content area is titled 'Case: CIP-B102-Lab4' and 'ADD A NEW HOST'. It contains four numbered instructions for adding a host, each with a corresponding input field:

- 1. Host Name:** The name of the computer being investigated. It can contain only letters, numbers, and symbols.
- 2. Description:** An optional one-line description or note about this computer.
- 3. Time zone:** An optional timezone value (i.e. EST5EDT). If not given, defaults to the local setting. A list of time zones can be found in the files.
- 4. Timeskew Adjustment:** An optional value to describe how many seconds this computer's clock was out of sync. For example, if the computer was 10 seconds fast, then enter -10 to compensate.

Fig 6.1.2 Image file details and integrity verification in Autopsy, specifying MD5 hash `d08a3881fc9bab83d60d5a8900f4e82c` and FAT32 partition settings.



The screenshot shows the 'Image File Details' and 'File System Details' sections of the Autopsy web interface. The browser window is the same as in Fig 6.1.1.

Image File Details

Local Name: images/CIPB102_Lab4_Ifeanyi_Raphael_USB.dd

Data Integrity: An MD5 hash can be used to verify the integrity of the image. (With split images, this hash is for the full image file)

- ☐ Ignore the hash value for this image.
- ☐ Calculate the hash value for this image.
- ☒ Add the following MD5 hash value for this image:

☐ Verify hash after importing?

File System Details

Analysis of the image file shows the following partitions:

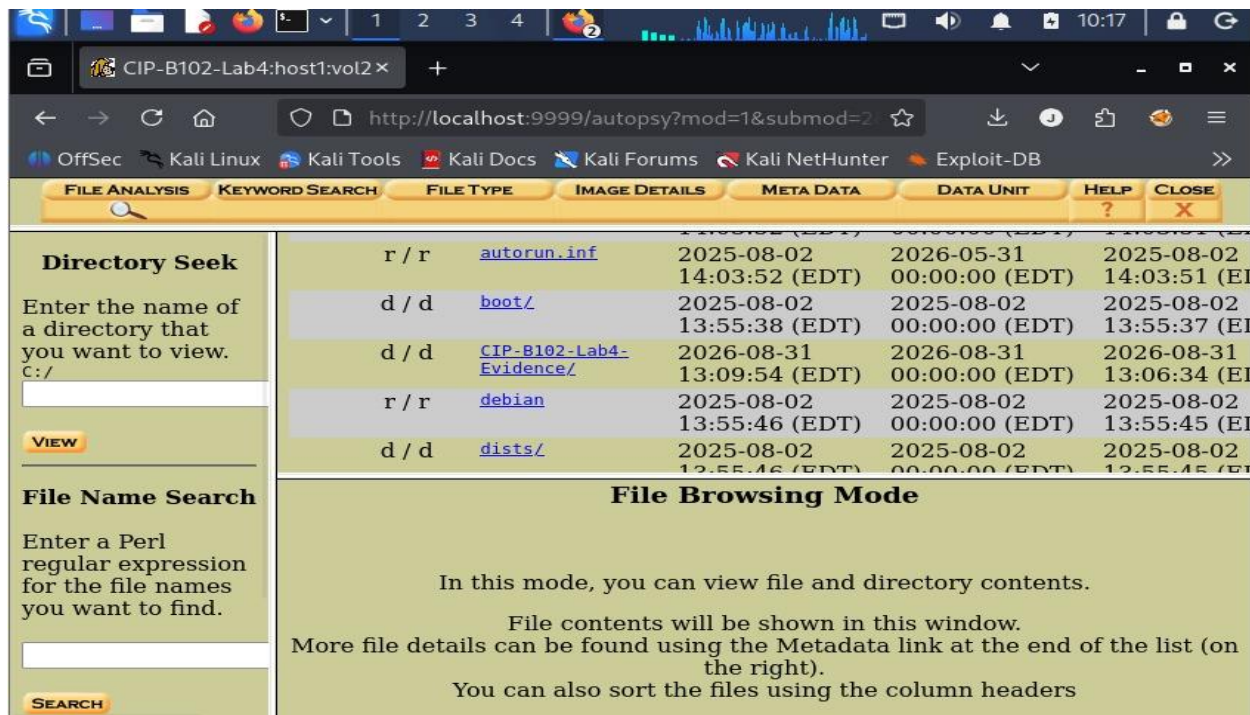
Partition 1 (Type: Win95 FAT32 (0x0c))

Add to case? ☒

Sector Range: 2048 to 13942687

Mount Point: File System Type:

Fig 6.1.3 Autopsy Directory Analysis Interface. Browsing root filesystem directories (C:/) to locate evidence artifacts and system files within the mounted FAT32 partition.

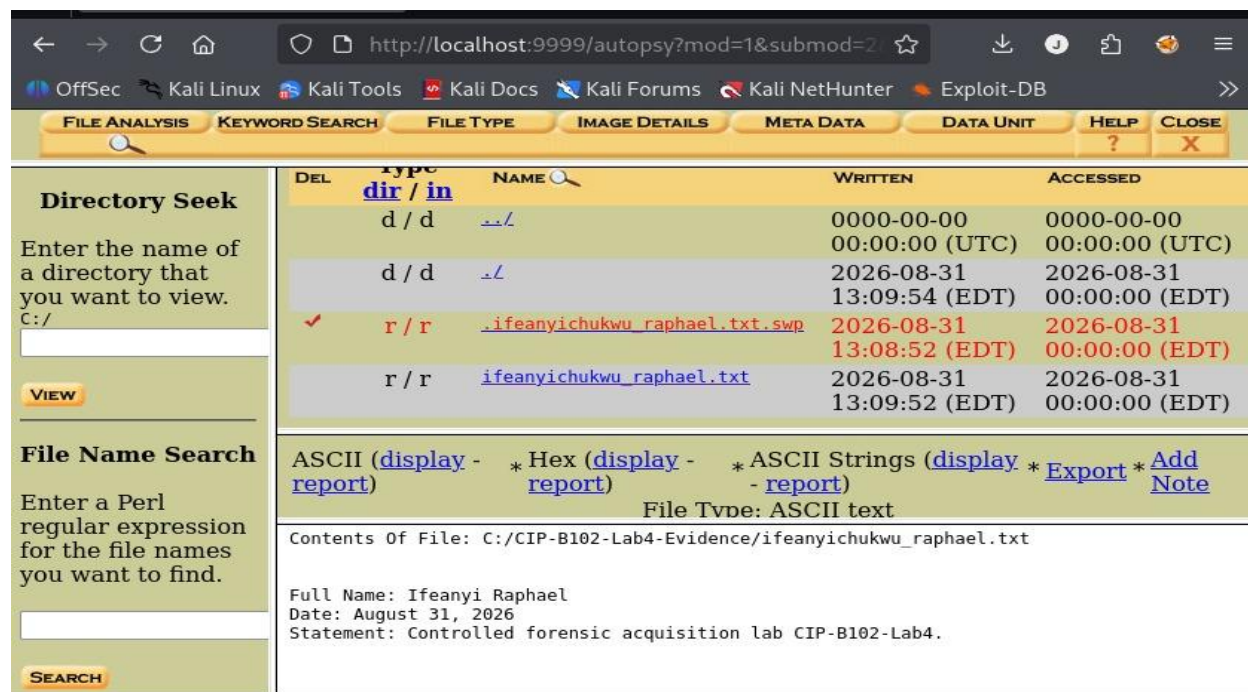


6.2 Directory Browse & Target Evidence Recovery

Navigating to the File Analysis module under C:/ revealed root directory entries and unallocated cluster records:

- Controlled Evidence Directory: Verified presence of CIP-B102-Lab4-Evidence/
- Evidence Text Artifact: Located Ifeanyichukwu_Raphael.txt (Inode 11).
- Extracted Forensic Artifacts: Recovered deleted spreadsheet Income.xls (Inode 13, 13,824 bytes, Author: Amelia Phillips) and surrounding text artifacts (confirmation.txt, letter1.txt, Regrets.doc).

Fig 6.2.0 Autopsy File Content & Evidence Verification Interface. Extracting ASCII contents of C:/CIP-B102-Lab4-Evidence/ifeanyichukwu_raphael.txt to confirm data integrity post-acquisition.



7. Technical Reflection & Low-Level Tool Risk Assessment

The use of low-level imaging tools like dd presents inherent operational risks that digital forensic examiners must manage:

1. **Critical Syntax Inversion (Destructive Overwrite Risk):**
The dd utility executes byte-level block commands without safety prompt confirmation. Reversing the input file (if=) and output file (of=) parameters will write destination data directly over original physical evidence, resulting in permanent, irreversible data destruction.
2. **Block Device Misidentification:**
Targeting an incorrect block node (e.g., selecting OS disk /dev/sda instead of target USB /dev/sdb) will overwrite host operating system volumes, crashing the system and destroying investigative tools.
3. **Write-Blocker Operational Mandate:**
Executing read operations directly on block storage without hardware write-blockers or

kernel-level read-only flags risks modifying file access timestamps (ATIME), dirty bits, or volume journal logs.

8. Conclusion & Chain of Custody Sign-Off

The forensic acquisition and validation procedure for Lab CIP-B102-Lab4 was successfully executed in full compliance with forensic standards. The acquired raw image CIPB102_Lab4_Ifeanyi_Raphael_USB.dd was cryptographically verified via matching MD5 and SHA-256 hashes, confirming exact bit-for-bit fidelity with physical USB block device /dev/sdb. Subsequent ingestion and artifact parsing within Autopsy validated that the target evidence file ifeanyichukwu_raphael.txt and all associated directory structures remain intact and admissible for formal forensic examination.